

Career AI Explorer

Project Outline

Honest career guidance for an AI-shaped future



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Section colors recur throughout the document so you can navigate visually:

	Problem & Quality		People & Closing		Product & Polish		Methodology & Cost		Data Sources
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Career advice still says *follow your passion* — but no one is honestly telling students which passions are walking into a wall.

- High schoolers choose what to study without knowing which fields are most exposed to AI displacement.
- **73%** of paralegal tasks are LLM-automatable (Felten 2023).
- Bookkeeper, paralegal, translator are top 5 declining roles (WEF 2025).
- AI capability has roughly doubled annually since 2022 (Stanford HAI).
- Generic career-test tools score on personality, not exposure to automation.

One user in mind:

- A Grade 11 student deciding what to study.
- Doesn't necessarily know what AI displacement means.
- Wants an honest read on the future they're walking into.
- Needs a path forward — not just doom.
- Will accept hard truths if they come with adjacent options and concrete next steps.

Constellation entry page

- Radial visualization of **1,016 occupations across 64 industries**.
- Browse by clicking any industry node, or take a 30-second matching quiz.
- Industries color-coded: azure (analytical), violet (people & care), mint (hands-on / safer), coral (higher AI exposure).

Analysis page

- **0–100 AI displacement score** for any occupation, optionally adjusted for company.
- Eight-dimension breakdown with explicit scoring formula.
- Evidence ledger — every claim cites its source (O*NET task, SEC filing, news article).
- Live news context — what's happening to this role in the last 60 days.
- Five adjacent careers ranked by skill overlap and resilience.
- Four to five concrete actions a Grade 11 student can take in the next 12 months.

News briefing

- **17 trusted news sources** (BBC, NYT, FT, Bloomberg, MIT Tech Review, IEEE Spectrum, etc.).

- Sources categorized by trust tier (Tier 1: research-grade · Tier 2: major outlets · Tier 3: aggregators).
- AI-categorized by lens: threat / tools / hiring / policy / macro.

Eight dimensions, weighted formula

Dimension	Weight	Direction
Routine cognitive	1.5	Increases score
AI capability gap	1.5	Increases score
Creative judgment	1.0	Decreases score
Regulatory moat	1.2	Decreases score
Embodied work	1.3	Decreases score
Stakes of error	1.0	Decreases score
Relationships	1.1	Decreases score
Physical dexterity	0.8	Decreases score

Aggregation formula

```
raw_up = 1.5 × routine + 1.5 × ai_gap (range 0–30)
raw_down = 1.0 × creative + 1.2 × regulatory + 1.3 × embodied
+ 1.0 × stakes + 1.1 × relationships + 0.8 × dexterity (range 0–64)
score = round( (raw_up / 30 × 55) + (100 – raw_down / 64 × 45) )
```

Calibration anchors (24 sourced reference roles)

Role	Score	Role	Score	Role	Score
Welder	12	Surgeon	28	Accountant	65
Plumber	14	Dentist	30	Graphic Designer	68
Electrician	16	Construction Manager	32	Paralegal	74
Kindergarten Teacher	18	Lawyer	48	Translator	78
Therapist	22	Truck Driver	50	Bookkeeper	80
Robotics Engineer	24	Investment Banker	52	Telemarketer	86
Nurse	26	Software Developer	60	Data Entry Clerk	88

■ Low (≤30)
 ■ Mid (31–55)
 ■ High (56+)

Median anchor: 43 / 100 — most roles cluster mid-range. All anchors sourced from Felten et al. 2023, McKinsey 2024, BLS 2024, WEF 2025.

All sources are free and public:

Primary task & occupation data

- **O*NET** — official US Department of Labor task lists for every occupation (~1,016 roles).
- **SEC EDGAR** — company 10-K filings used for company-specific analysis when the user provides one.

News & current signals

- 17 trusted news RSS feeds (BBC, NYT, FT, Bloomberg, MIT Tech Review, IEEE Spectrum, ArXiv, etc.).
- Sources tiered by trust level so the analysis weights research-grade sources higher than aggregators.

Research & calibration

- Felten et al. 2023 — AI Occupational Exposure (52 capabilities × 800 roles).
- McKinsey 2024 — task-level automation potential.
- BLS 2024 — Occupational Employment Statistics + 2024–34 employment projections.
- WEF 2025 — Future of Jobs Report.
- Anthropic Economic Index, Stanford HAI 2026 AI Index — capability benchmarks.
- Wikipedia & ArXiv — public summaries and primary research papers.

Frontend

- Three HTML pages: **constellation.html**, **mockup.html** (analysis), **news.html**.
- Ocean Depths design system: azure, mint, violet, coral on a deep navy gradient.
- Fraunces variable serif typography for headlines with italic gradient accent words.
- Glass-morphism cards (translucent + backdrop-filter blur) with hairline borders and inset highlights.
- Aurora swoosh ribbons and animated nebula backgrounds in both dark and light themes.

Backend

- Express.js server with provider abstraction layer.
- Provider routing in **auto** mode: Demo provider → Local Ollama (Llama 3.2 3B) → Anthropic Claude as fallback.
- Disk-cache layer for analyzed occupations — repeat searches are instant.

Always-on infrastructure

- Server runs as a macOS LaunchAgent — auto-starts on login, restarts on crash.

- Cache-warmer LaunchAgent runs hourly to pre-populate popular occupations.
- Native macOS desktop app via Electron with hot-reload from source.
- File watcher (chokidar) auto-reloads window on edits to HTML or server JS within ~200 ms.

Layer	Cost	When it runs
Demo provider (10 popular roles)	\$0	Hand-crafted JSON served instantly
Disk cache	\$0	Any repeat search hits the cache
Local Ollama (Llama 3.2 3B)	\$0	Any new occupation, runs on user's Mac
Anthropic Claude (fallback only)	Paid	Never required — only if user opts in

- Local LLM runs on the user's Mac — no API fees.
- 10 popular roles served instantly from hand-crafted demo cache.
- Cache layer makes every repeat search instant.
- **Total monthly cost: \$0.**

#	Feature	How it works
F1	Risk-color score halo	mint / azure / coral by score tier
F2	Animated mesh gradient background 30s drift cycle	
F3	Compact tabular dimension grid	Cards / Table toggle, persisted
F4	Variable-font hover transitions	Fraunces weight 400→600 + opsz 24→96
F5	Drop cap on score caption	::first-letter Fraunces italic 64px
F6	Tabular numerals on all stats	font-variant-numeric: tabular-nums
F7	Gradient text on italic accent words	azure → mint → violet text-fill
F8	Magnetic industry nodes	lean toward cursor within 60px
F9	Slot-machine score number morph	digit strips translateY
F10	Constellation entrance animation	industries fly in + line draw-in
F11	3D parallax tilt on hero glass	scroll-tied rotateX
F12	Cursor trail on constellation	8–12 pooled fading dots
F13	Section reveal wipe	clip-path inset sweep
F14	Light-source-aware glass	cursor-follow radial highlight
F15	Frosted noise texture	2% SVG noise overlay on glass
F16	Edge glow on hovered cards	inset 8px azure + outer glow
F17	Layered constellation rings parallax	3 rings at different speeds

Loading & states

- Smart loader with network-graph SVG, smart-paced steps, indeterminate "Synthesizing" anchor, elapsed time counter.
- Skeleton glass cards during loading.
- Polished error card with retry button + collapsible technical details.
- Empty-state hint on constellation: "Hover an industry to explore".

Navigation & controls

- Dark / light theme toggle with localStorage persistence.
- Cmd+K command palette — instant hover, no backdrop-filter lag.
- Cross-page view transitions — smooth fade between constellation, analysis, and news.

Cross-page continuity

- Persistent backdrop blob phase across page navigation (sessionStorage-backed).
- Glass status bar on every page showing live server health.
- Industry color legend, live-stat rotating panel (8 sourced facts), aurora swooshes — appearing identically across all pages so the three pages feel like one product.

I designed

- The 8-dimension scoring framework and weighting.
- All 24 calibration score anchors and their sources.
- The trusted-news-source policy (which 17 sources, what trust tier).
- The constellation visualization and matching quiz logic.
- The evidence ledger schema (every claim must cite a source).
- The methodology page itself and the "hopeful path" framing.

AI's role

- Synthesizes the curated data into structured analysis at runtime.
- Helped with code generation — the same way a calculator helps a math student.
- Never decided what dimensions matter, what sources to trust, or how to score.

Every substantive design and content decision is mine. AI is a feature *of* the tool — it analyzes data — and AI was a coding assistant.

Find your future.

1,016 occupations · 64 industries · live data · zero cost · honest scoring

